Saeid Pourmand

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Research Interests

Mathematical Foundations of Efficient Deep Learning; Low-Precision and Binary Neural Networks; Randomized Representations and Similarity Estimation; Geometry and Optimization of Neural Networks; Model Quantization and Compression; Efficient Large Language Models

Education

Ph.D. in Artificial Intelligence, Oregon State University, USA *Sep 2023 - Present*

Advisor: Alireza Aghasi, Expected Graduation Date: June 2027

M.Sc. in Mathematical Statistics, National University of Iran, Tehran, Iran *Sep 2017 - Feb 2020*

B.Sc. in Statistics, University of Guilan, Rasht, Iran *Sep 2013 - Jun 2017*

Publications

[1] Ahmed, A., Pourmand, S., Awais, M., Farooq, R., Aghasi, A., Angular Networks: Low-Bit Learning from Randomized Similarity Estimators *Advances in Neural Information Processing System*, To Appear (2026)

[2] Aghasi, A., Marshall, N. F., Pourmand, S., and Whiting, W. G-Net: A Provably Easy Construction of High-Accuracy Random Binary Neural Networks. *Advances in Neural Information Processing Systems*, 38, 154658-154716 (2025). [Code]

[3] Pourmand, S., Whiting, W. D., Aghasi, A., and Marshall, N. F. Laplace-hdc: Understanding the geometry of binary hyperdimensional computing. *Journal of Artificial Intelligence Research*, 82, 1293-1323 (2025).[Code]

[4] Pourmand, S., Shabbak, A., and Ganjali, M. Feature Selection Based on Divergence Functions: A Comparative Classification Study. *Journal of Statistics, Optimization and Information Computing*, 9(3) (2021): 587-606.

[5] Khanjari, A., Pourmand, S., and FaridRouhani, M. R., Manually Selecting The Data Function for Supervised Learning of small datasets. *Arxiv Preprint*, arXiv:2303.03894 (2023).

Research Experience

Post-Training Quantization of Large Language Models (Ongoing)

- Independently initiated and currently lead research investigating randomized transformations for post-training quantization of large language models.

Mathematical Investigations in Randomized Binary Embeddings 2026

- Independently investigated variance-reduction techniques for randomized binary embeddings, deriving exact variance expressions for stratified estimators.
- Explored alternative binary random-feature mappings through probabilistic analysis and empirical evaluation, examining their mathematical properties and limitations.

Angular Networks - Low-Bit Learning via Angular Geometry 2025-2026

- Conceived the central idea behind Angular Networks, a low-precision learning framework based on randomized estimators of angular similarity rather than conventional Euclidean inner-product approximations.
- Designed and led the experimental investigation, including baseline selection, implementation, and

evaluation across CNNs and Transformers on vision, language, and medical time-series benchmarks.
- Demonstrated strong low-bit learning performance across multiple architectures and datasets, including ImageNet-1K.

G-Net - Randomized Binary Neural Network Construction 2024-2025
- Designed and conducted experimental evaluations of G-Net, a framework for embedding full-precision neural networks into high-dimensional binary representations without retraining or fine-tuning.
- Selected baselines and benchmarked G-Net across image, time-series, and language classification datasets, achieving approximately 81% accuracy on CIFAR-10.
- Empirically investigated Rademacher projections as alternatives to Gaussian random projections, demonstrating comparable performance across different embedding dimensions.

Laplace-HDC - Geometry of Binary Hyperdimensional Computing 2023-2024
- Designed and conducted experimental evaluations of Laplace-HDC, a randomized binary embedding framework that preserves similarity structures.
- Selected and implemented comparison baselines, benchmarking Laplace-HDC against existing hyperdimensional computing methods on MNIST and Fashion-MNIST, with reported improvements of 2% and 4%, respectively.

Divergence-Based Feature Selection - Statistical Learning and Classification 2020-2021
- Conceived and led a research project investigating divergence-based feature selection, exploring how differences between probability distributions can be used to identify informative features for classification.
- Designed and conducted a comprehensive experimental study comparing divergence-based feature selection methods across 21 benchmark datasets, evaluating classification accuracy, F1-score, and feature selection effectiveness.
- Authored the manuscript, synthesizing the mathematical formulations, experimental methodology, and comparative findings into a first-author peer-reviewed journal publication.

Mentorship and Academic Services

- **Research Co-Mentor**, Oregon State University 2025-2026
  Co-mentored a master's student and an undergraduate researcher on the *Angular Networks* project, guiding research and software development jointly with my Ph.D. advisor.

Honors & Awards

- Edith McDougall Scholarship, Oregon State University 2024
- Edith McDougall Scholarship, Oregon State University 2023
- Outstanding Student Award, National University of Iran 2020
- Highest cumulative GPA in Master's and Bachelor's programs 2017, 2020

Technical Skills

Programming: Python, R, MATLAB
Frameworks & Tools: PyTorch, NumPy, Transformers, CUDA, HPC, Git, Hugging Face, Linux, Conda, Latex

Conference Presentations

G-Net (NeurIPS 2025, poster)